

Coordinate Geometry

ANSWER KEY



Distance and midpoint formulas

- 1 Find the distance between $(2, -1)$ and $(10, 5)$.

$$\text{Distance} = \sqrt{(10 - 2)^2 + (5 - (-1))^2} = \sqrt{64 + 36} = \sqrt{100} = 10.$$

- 2 Find the midpoint of the segment connecting $(-4, 7)$ and $(6, -3)$.

$$\text{Midpoint} = \left(\frac{-4 + 6}{2}, \frac{7 + (-3)}{2} \right) = (1, 2).$$

Slope and equations of lines

- 3 Find the slope of the line through $(1, 4)$ and $(5, -8)$.

$$\text{Slope} = \frac{-8 - 4}{5 - 1} = \frac{-12}{4} = -3.$$

- 4 Write the equation of the line through $(2, -3)$ with slope $\frac{1}{2}$, in slope-intercept form.

$$y - (-3) = \frac{1}{2}(x - 2), \text{ so } y + 3 = \frac{1}{2}x - 1, \text{ giving } y = \frac{1}{2}x - 4.$$

Parallel and perpendicular lines

- 5 Find the equation of the line parallel to $y = 4x - 1$ passing through $(0, 6)$.

Parallel lines share the same slope: $m = 4$. Since $(0, 6)$ is the y -intercept, the equation is $y = 4x + 6$.

- 6 Find the equation of the line perpendicular to $y = -\frac{2}{3}x + 5$ passing through $(4, 1)$.

The perpendicular slope is the negative reciprocal: $m = \frac{3}{2}$. $y - 1 = \frac{3}{2}(x - 4)$, so
 $y = \frac{3}{2}x - 6 + 1 = \frac{3}{2}x - 5$.

The equation of a circle

- 7 Write the equation of a circle with center $(3, -2)$ and radius 7.

$$(x - 3)^2 + (y + 2)^2 = 49.$$

- 8 Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 3 = 0$, by completing the square.

Group and complete the square: $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 3 + 9 + 4$, giving
 $(x - 3)^2 + (y + 2)^2 = 16$. Center: $(3, -2)$. Radius: $\sqrt{16} = 4$.

Reflections, translations, and symmetry

- 9 Find the coordinates of the point $(5, -3)$ after a reflection over the x -axis, then after a reflection over the y -axis.

Reflection over the x -axis: $(5, 3)$ (the y -coordinate's sign flips). Reflection over the y -axis (applied to the original point): $(-5, -3)$ (the x -coordinate's sign flips).

- 10 A point (a, b) is translated 3 units right and 5 units down. Write the coordinates of the image point.
 $(a + 3, b - 5)$.

Finding area using the coordinate plane

- 11 Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$, and $(0, 4)$.

Since the triangle has a right angle at the origin, with legs along the axes: $\text{Area} = \frac{1}{2}(6)(4) = 12$.

- 12 A rectangle has vertices $(1, 1)$, $(1, 5)$, $(7, 5)$, and $(7, 1)$. Find its area and perimeter.

Width: $7 - 1 = 6$. Height: $5 - 1 = 4$. $\text{Area} = 6 \times 4 = 24$. $\text{Perimeter} = 2(6 + 4) = 20$.

Determining if points are collinear

- 13 Determine whether the points $(1, 2)$, $(3, 6)$, and $(5, 10)$ are collinear, by comparing slopes.

Slope from $(1, 2)$ to $(3, 6)$: $\frac{6 - 2}{3 - 1} = \frac{4}{2} = 2$. Slope from $(3, 6)$ to $(5, 10)$: $\frac{10 - 6}{5 - 3} = \frac{4}{2} = 2$. Since both slopes are equal, the three points are collinear.

Finding the intersection of two lines

- 14 Find the point of intersection of $y = 2x - 3$ and $y = -x + 9$.

Set equal: $2x - 3 = -x + 9$, so $3x = 12$, giving $x = 4$. Then $y = 2(4) - 3 = 5$. Intersection: $(4, 5)$.

Combining coordinate geometry concepts

- 15** This question has two parts. Triangle ABC has vertices $A(0, 0)$, $B(4, 0)$, and $C(2, 6)$. (a) Find the length of each side of the triangle. (b) Determine whether the triangle is isosceles, and justify your answer.

(a) $AB = \sqrt{(4 - 0)^2 + (0 - 0)^2} = 4$. $AC = \sqrt{(2 - 0)^2 + (6 - 0)^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$.
 $BC = \sqrt{(2 - 4)^2 + (6 - 0)^2} = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}$. (b) Since $AC = BC = 2\sqrt{10}$ (two sides are equal in length), the triangle is isosceles.

Graphing inequalities in the coordinate plane

- 16** Determine whether the point $(3, 1)$ satisfies the inequality $y < 2x - 3$.

Substitute: $1 < 2(3) - 3 = 3$? True. The point satisfies the inequality.

- 17** A region is defined by $y \geq x + 1$. Determine whether $(0, 0)$ lies in this region.

Substitute: $0 \geq 0 + 1 = 1$? False. The point $(0, 0)$ does not lie in the region.

The equation of a parabola in the coordinate plane

- 18** Write the equation of a parabola with vertex $(2, -3)$ that passes through $(4, 5)$.

Vertex form: $y = a(x - 2)^2 - 3$. Substitute $(4, 5)$: $5 = a(4 - 2)^2 - 3$, so $5 = 4a - 3$, giving $a = 2$.
Equation: $y = 2(x - 2)^2 - 3$.

Finding points that divide a segment in a given ratio

- 19** Find the point that divides the segment from $(1, 2)$ to $(9, 10)$ in a ratio of $1 : 3$ (closer to the first point).

Using the section formula for a $1 : 3$ ratio: $x = 1 + \frac{1}{4}(9 - 1) = 1 + 2 = 3$. $y = 2 + \frac{1}{4}(10 - 2) = 2 + 2 = 4$.
Point: $(3, 4)$.

A multi-part problem combining lines and circles

- 20** This question has two parts. A circle has equation $(x - 2)^2 + (y + 1)^2 = 25$. (a) State the center and radius. (b) Determine whether the point $(5, 3)$ lies inside, on, or outside the circle, by computing its distance from the center.

(a) Center: $(2, -1)$. Radius: $\sqrt{25} = 5$. (b) Distance from center to $(5, 3)$:
 $\sqrt{(5 - 2)^2 + (3 - (-1))^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Since this distance equals the radius, the point $(5, 3)$ lies exactly ON the circle.

Systems of a line and a circle

- 21 Find the points of intersection of the line $y = x + 1$ and the circle $x^2 + y^2 = 25$.

Substitute: $x^2 + (x + 1)^2 = 25$, so $x^2 + x^2 + 2x + 1 = 25$, giving $2x^2 + 2x - 24 = 0$, or $x^2 + x - 12 = 0$. Factor: $(x + 4)(x - 3) = 0$, so $x = -4$ or $x = 3$. Corresponding y -values: for $x = -4$, $y = -3$; for $x = 3$, $y = 4$. Intersection points: $(-4, -3)$ and $(3, 4)$.

Multiple choice — easy

- 22 Find the distance between $(8, 8)$ and $(18, 32)$.

- a. 27
- b. 34
- c. 25
- d. 26 ✓

$$\text{Distance} = \sqrt{(18 - 8)^2 + (32 - 8)^2} = \sqrt{(10)^2 + (24)^2} = \sqrt{676} = 26.$$

- 23 Find the distance between $(1, -2)$ and $(-26, 118)$.

- a. 123 ✓
- b. 122
- c. 124
- d. 147

$$\text{Distance} = \sqrt{(-26 - 1)^2 + (118 - (-2))^2} = \sqrt{(-27)^2 + (120)^2} = \sqrt{15129} = 123.$$

- 24 Find the distance between $(-2, 7)$ and $(-29, -29)$.

- a. 63
- b. 46
- c. 45 ✓
- d. 44

$$\text{Distance} = \sqrt{(-29 - (-2))^2 + (-29 - 7)^2} = \sqrt{(-27)^2 + (-36)^2} = \sqrt{2025} = 45.$$

25 Find the distance between $(-5, 0)$ and $(4, 40)$.

- a. 49
- b. 42
- c. 41 ✓
- d. 40

$$\text{Distance} = \sqrt{(4 - (-5))^2 + (40 - (0))^2} = \sqrt{(9)^2 + (40)^2} = \sqrt{1681} = 41.$$

26 Find the distance between $(0, 2)$ and $(8, 17)$.

- a. 16
- b. 18
- c. 17 ✓
- d. 23

$$\text{Distance} = \sqrt{(8 - (0))^2 + (17 - (2))^2} = \sqrt{(8)^2 + (15)^2} = \sqrt{289} = 17.$$

27 Find the distance between $(5, 8)$ and $(-31, -40)$.

- a. 60 ✓
- b. 61
- c. 84
- d. 59

$$\text{Distance} = \sqrt{(-31 - (5))^2 + (-40 - (8))^2} = \sqrt{(-36)^2 + (-48)^2} = \sqrt{3600} = 60.$$

28 Find the distance between $(-2, 4)$ and $(5, -20)$.

- a. 26
- b. 24
- c. 31
- d. 25 ✓

$$\text{Distance} = \sqrt{(5 - (-2))^2 + (-20 - (4))^2} = \sqrt{(7)^2 + (-24)^2} = \sqrt{625} = 25.$$

29 Find the distance between $(-4, -7)$ and $(6, -31)$.

a. 26 ✓

b. 34

c. 25

d. 27

$$\text{Distance} = \sqrt{(6 - (-4))^2 + (-31 - (-7))^2} = \sqrt{(10)^2 + (-24)^2} = \sqrt{676} = 26.$$

30 Find the distance between $(6, -3)$ and $(-12, -27)$.

a. 29

b. 42

c. 30 ✓

d. 31

$$\text{Distance} = \sqrt{(-12 - (6))^2 + (-27 - (-3))^2} = \sqrt{(-18)^2 + (-24)^2} = \sqrt{900} = 30.$$

31 Find the distance between $(1, -7)$ and $(10, -47)$.

a. 49

b. 40

c. 42

d. 41 ✓

$$\text{Distance} = \sqrt{(10 - (1))^2 + (-47 - (-7))^2} = \sqrt{(9)^2 + (-40)^2} = \sqrt{1681} = 41.$$

32 Find the distance between $(-7, -6)$ and $(17, -51)$.

- a. 52
- b. 50
- c. 51 ✓
- d. 69

$$\text{Distance} = \sqrt{(17 - (-7))^2 + (-51 - (-6))^2} = \sqrt{(24)^2 + (-45)^2} = \sqrt{2601} = 51.$$

33 Find the distance between $(-7, -2)$ and $(-25, -26)$.

- a. 29
- b. 30 ✓
- c. 31
- d. 42

$$\text{Distance} = \sqrt{(-25 - (-7))^2 + (-26 - (-2))^2} = \sqrt{(-18)^2 + (-24)^2} = \sqrt{900} = 30.$$

34 Find the distance between $(3, -5)$ and $(-17, 43)$.

- a. 52 ✓
- b. 53
- c. 68
- d. 51

$$\text{Distance} = \sqrt{(-17 - (3))^2 + (43 - (-5))^2} = \sqrt{(-20)^2 + (48)^2} = \sqrt{2704} = 52.$$

35 Find the distance between $(-8, -6)$ and $(-3, 6)$.

- a. 13 ✓
- b. 14
- c. 17
- d. 12

$$\text{Distance} = \sqrt{(-3 - (-8))^2 + (6 - (-6))^2} = \sqrt{(5)^2 + (12)^2} = \sqrt{169} = 13.$$

36 Find the distance between $(8, -5)$ and $(-19, -125)$.

a. 122

b. 123 ✓

c. 124

d. 147

$$\text{Distance} = \sqrt{(-19 - 8)^2 + (-125 - (-5))^2} = \sqrt{(-27)^2 + (-120)^2} = \sqrt{15129} = 123.$$

Multiple choice — medium

37 Find the midpoint of the segment connecting $(-9, -10)$ and $(1, -8)$.

a. $(4, -9)$

b. $(-4, -9)$ ✓

c. $(-4, -8)$

d. $(-3, -9)$

$$\text{Midpoint} = \left(\frac{-9 + 1}{2}, \frac{-10 - 8}{2} \right) = (-4, -9).$$

38 Find the midpoint of the segment connecting $(9, -5)$ and $(1, 11)$.

a. $(5, 4)$

b. $(6, 3)$

c. $(5, 3)$ ✓

d. $(-5, 3)$

$$\text{Midpoint} = \left(\frac{9 + 1}{2}, \frac{-5 + 11}{2} \right) = (5, 3).$$

39 Find the midpoint of the segment connecting (1, 9) and (−7, 9).

- a. (−2, 9)
- b. (3, 9)
- c. (−3, 9) ✓
- d. (−3, 10)

$$\text{Midpoint} = \left(\frac{1 - 7}{2}, \frac{9 + 9}{2} \right) = (-3, 9).$$

40 Find the midpoint of the segment connecting (−6, 0) and (4, 12).

- a. (1, 6)
- b. (0, 6)
- c. (−1, 6) ✓
- d. (−1, 7)

$$\text{Midpoint} = \left(\frac{-6 + 4}{2}, \frac{0 + 12}{2} \right) = (-1, 6).$$

41 Find the midpoint of the segment connecting (8, −8) and (−4, 0).

- a. (2, −3)
- b. (2, −4) ✓
- c. (3, −4)
- d. (−2, −4)

$$\text{Midpoint} = \left(\frac{8 - 4}{2}, \frac{-8}{2} \right) = (2, -4).$$

42 Find the midpoint of the segment connecting (2, 8) and (10, −4).

- a. (6, 3)
- b. (6, 2) ✓
- c. (7, 2)
- d. (−6, 2)

$$\text{Midpoint} = \left(\frac{2 + 10}{2}, \frac{8 - 4}{2} \right) = (6, 2).$$

43 Find the midpoint of the segment connecting $(-3, 1)$ and $(9, 11)$.

a. $(3, 6)$ ✓

b. $(3, 7)$

c. $(4, 6)$

d. $(-3, 6)$

$$\text{Midpoint} = \left(\frac{-3 + 9}{2}, \frac{1 + 11}{2} \right) = (3, 6).$$

44 Find the midpoint of the segment connecting $(7, 3)$ and $(-11, -11)$.

a. $(-2, -3)$

b. $(-1, -4)$

c. $(2, -4)$

d. $(-2, -4)$ ✓

$$\text{Midpoint} = \left(\frac{7 - 11}{2}, \frac{3 - 11}{2} \right) = (-2, -4).$$

45 Find the midpoint of the segment connecting $(-7, -5)$ and $(3, 3)$.

a. $(-2, 0)$

b. $(2, -1)$

c. $(-1, -1)$

d. $(-2, -1)$ ✓

$$\text{Midpoint} = \left(\frac{-7 + 3}{2}, \frac{-5 + 3}{2} \right) = (-2, -1).$$

46 Find the midpoint of the segment connecting $(8, 1)$ and $(0, -7)$.

a. $(5, -3)$

b. $(4, -3)$ ✓

c. $(4, -2)$

d. $(-4, -3)$

$$\text{Midpoint} = \left(\frac{8+0}{2}, \frac{1+(-7)}{2} \right) = (4, -3).$$

47 Find the midpoint of the segment connecting $(4, 4)$ and $(-8, -10)$.

a. $(2, -3)$

b. $(-2, -2)$

c. $(-2, -3)$ ✓

d. $(-1, -3)$

$$\text{Midpoint} = \left(\frac{4+(-8)}{2}, \frac{4+(-10)}{2} \right) = (-2, -3).$$

48 Find the midpoint of the segment connecting $(-7, -3)$ and $(3, 1)$.

a. $(-2, 0)$

b. $(-2, -1)$ ✓

c. $(2, -1)$

d. $(-1, -1)$

$$\text{Midpoint} = \left(\frac{-7+3}{2}, \frac{-3+1}{2} \right) = (-2, -1).$$

49 Find the midpoint of the segment connecting $(2, -5)$ and $(-2, 5)$.

a. $(1, 0)$

b. $(1, -2)$

c. $(0, 0)$ ✓

d. $(0, 1)$

$$\text{Midpoint} = \left(\frac{2+(-2)}{2}, \frac{-5+5}{2} \right) = (0, 0).$$

50 Find the midpoint of the segment connecting $(12, 9)$ and $(-4, 11)$.

a. $(-4, 10)$

b. $(4, 11)$

c. $(5, 10)$

d. $(4, 10)$ ✓

$$\text{Midpoint} = \left(\frac{12 - 4}{2}, \frac{9 + 11}{2} \right) = (4, 10).$$

51 Find the midpoint of the segment connecting $(8, 7)$ and $(-8, -9)$.

a. $(1, -1)$

b. $(0, -1)$ ✓

c. $(0, 0)$

d. $(-2, 0)$

$$\text{Midpoint} = \left(\frac{8 - 8}{2}, \frac{7 - 9}{2} \right) = (0, -1).$$

Multiple choice — hard

52 Find the radius of the circle $x^2 + y^2 + 16x - 18y + 81 = 0$, by completing the square.

a. 8 ✓

b. 7

c. 64

d. 9

Completing the square: $(x + 8)^2 + (y - 9)^2 = 64$. Center: $(-8, 9)$. Radius: $\sqrt{64} = 8$.

53 Find the radius of the circle $x^2 + y^2 - 18x - 16y + 141 = 0$, by completing the square.

a. 3

b. 1

c. 4

d. 2 ✓

Completing the square: $(x - 9)^2 + (y - 8)^2 = 4$. Center: (9, 8). Radius: $\sqrt{4} = 2$.

54 Find the radius of the circle $x^2 + y^2 - 16y + 15 = 0$, by completing the square.

a. 49

b. 7 ✓

c. 8

d. 6

Completing the square: $(x)^2 + (y - 8)^2 = 49$. Center: (0, 8). Radius: $\sqrt{49} = 7$.

55 Find the radius of the circle $x^2 + y^2 + 10x - 4y + 4 = 0$, by completing the square.

a. 25

b. 5 ✓

c. 6

d. 4

Completing the square: $(x + 5)^2 + (y - 2)^2 = 25$. Center: (-5, 2). Radius: $\sqrt{25} = 5$.

56 Find the radius of the circle $x^2 + y^2 - 8x - 16y - 41 = 0$, by completing the square.

a. 11 ✓

b. 121

c. 12

d. 10

Completing the square: $(x - 4)^2 + (y - 8)^2 = 121$. Center: (4, 8). Radius: $\sqrt{121} = 11$.

57 Find the equation of the line perpendicular to $y = -\frac{5}{4}x$ passing through $(0, -3)$, in slope-intercept form.

a. $y = -\frac{4}{5}x - 3$

b. $y = \frac{4}{5}x - 3$ ✓

c. $y = -\frac{5}{4}x - 3$

d. $y = \frac{4}{5}x - 2$

The perpendicular slope is the negative reciprocal: $m = \frac{4}{5}$. $y - (-3) = \frac{4}{5}(x - 0)$, giving $y = \frac{4}{5}x - 3$.

58 Find the equation of the line perpendicular to $y = -\frac{1}{4}x$ passing through $(0, 5)$, in slope-intercept form.

a. $y = -\frac{1}{4}x + 5$

b. $y = 4x + 6$

c. $y = -4x + 5$

d. $y = 4x + 5$ ✓

The perpendicular slope is the negative reciprocal: $m = 4$. $y - (5) = 4(x - 0)$, giving $y = 4x + 5$.

59 Find the equation of the line perpendicular to $y = \frac{1}{5}x$ passing through $(3, 7)$, in slope-intercept form.

a. $y = 5x + 22$

b. $y = \frac{1}{5}x + 22$

c. $y = -5x + 23$

d. $y = -5x + 22$ ✓

The perpendicular slope is the negative reciprocal: $m = -5$. $y - (7) = -5(x - 3)$, giving $y = -5x + 22$.

60 Find the equation of the line perpendicular to $y = -\frac{2}{3}x$ passing through $(3, -1)$, in slope-intercept form.

a. $y = \frac{3}{2}x - \frac{9}{2}$

b. $y = -\frac{2}{3}x - \frac{11}{2}$

c. $y = -\frac{3}{2}x - \frac{11}{2}$

d. $y = \frac{3}{2}x - \frac{11}{2}$ ✓

The perpendicular slope is the negative reciprocal: $m = \frac{3}{2}$. $y - (-1) = \frac{3}{2}(x - 3)$, giving $y = \frac{3}{2}x - \frac{11}{2}$.

61 Find the equation of the line perpendicular to $y = \frac{5}{4}x$ passing through $(7, -3)$, in slope-intercept form.

a. $y = \frac{4}{5}x + \frac{13}{5}$

b. $y = -\frac{4}{5}x + \frac{18}{5}$

c. $y = -\frac{4}{5}x + \frac{13}{5}$ ✓

d. $y = \frac{5}{4}x + \frac{13}{5}$

The perpendicular slope is the negative reciprocal: $m = -\frac{4}{5}$. $y - (-3) = -\frac{4}{5}(x - 7)$, giving $y = -\frac{4}{5}x + \frac{13}{5}$.

62 Find the point of intersection of $y = 3x - 17$ and $y = -2x - 2$.

a. $(3, -8)$ ✓

b. $(4, -8)$

c. $(-3, -8)$

d. $(3, 8)$

Set equal: $3x - 17 = -2x - 2$, giving $x = 3$. Then $y = 3(3) - 17 = -8$. Intersection: $(3, -8)$.

63 Find the point of intersection of $y = -x + 5$ and $y = -2x + 1$.

a. $(4, 9)$

b. $(-4, 9)$ ✓

c. $(-3, 9)$

d. $(-4, -9)$

Set equal: $-x + 5 = -2x + 1$, giving $x = -4$. Then $y = -(-4) + 5 = 9$. Intersection: $(-4, 9)$.

64 Find the point of intersection of $y = x - 12$ and $y = 4x - 39$.

- a. $(9, 3)$
- b. $(-9, -3)$
- c. $(9, -3)$ ✓
- d. $(10, -3)$

Set equal: $x - 12 = 4x - 39$, giving $x = 9$. Then $y = (9) - 12 = -3$. Intersection: $(9, -3)$.

65 Find the point of intersection of $y = -x$ and $y = 2x - 6$.

- a. $(2, 2)$
- b. $(3, -2)$
- c. $(2, -2)$ ✓
- d. $(-2, -2)$

Set equal: $-x = 2x - 6$, giving $x = 2$. Then $y = -(2) = -2$. Intersection: $(2, -2)$.

66 Find the point of intersection of $y = 5x + 38$ and $y = -3x - 10$.

- a. $(-5, 8)$
- b. $(6, 8)$
- c. $(-6, -8)$
- d. $(-6, 8)$ ✓

Set equal: $5x + 38 = -3x - 10$, giving $x = -6$. Then $y = 5(-6) + 38 = 8$. Intersection: $(-6, 8)$.